2VG3 Homework 4 - RigidBody Collisions in 2D

Exercise 2 (a) - Collision Details with e=1.0

Scenario 0: Default setup with no initial rotation.

Collision 0: t=2.504166666666686, collision point=[-0.004171311855316162, 25.0, 0.0], normal=[1.0, 0.0, -0.0], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-0.016671180725097656, 25.0, -0.5) (body 1 corner against body 0 face); contact 2 at (0.008328557014465332, 25.0, 0.5) (body 0 corner against body 1 face).

Scenario 1: Body 0 starts rotated by 30 degrees.

Collision 0: t=2.441666666666683, collision point=[0.23332858085632324, 25.0, -0.14322692155838013], normal=[1.0, 0.0, -0.0], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (0.23332858085632324, 25.0, -0.1617315709590912) (face-face (edge-edge) contact); contact 2 at (0.23332858085632324, 25.0, -0.12472227215766907) (face-face (edge-edge) contact).

Scenario 2: Body 0 starts at 60 degrees and body 1 starts at 30 degrees.

Collision 0: t=2.475000000000018, collision point=[-0.2613295912742615, 25.0, 0.1339745968580246], normal=[0.8660253882408142, 0.0, 0.5], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-0.25897935032844543, 25.0, 0.1299038976430893) (face-face (edge-edge) contact); contact 2 at (-0.2636798024177551, 25.0, 0.1380452960729599) (face-face (edge-edge) contact).

Scenario 3: Body 0 starts spinning at -5 rad/s, while body 1 is moved to x=-1.9, z=1.9 and starts from rest.

Collision 0: t=1.6999999999999953, collision point=[-2.403679370880127, 25.0, 0.8999999761581421], normal=[-0.0, 0.0, 1.0], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-2.403160333633423, 25.0, 0.8999999761581421) (face-face (edge-edge) contact); contact 2 at (-2.404198408126831, 25.0, 0.8999999761581421) (face-face (edge-edge) contact).

Scenario 4: Body 0 starts spinning at +3 rad/s, and body 1 starts at -10 degrees.

Collision 0: t=2.5166666666666866, collision point=[0.12067129462957382, 25.0, -0.6252517104148865], normal=[0.9541522860527039, 0.0, 0.29932165145874023], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (0.13062013685703278, 25.0, -0.6569657921791077) (face-face (edge-edge) contact); contact 2 at (0.11072244495153427, 25.0, -0.5935376286506653) (face-face (edge-edge) contact).

Collision 1: t=2.5458333333333547, collision point=[-0.1615152209997177, 25.0, 0.5889225006103516], normal=[0.963695228099823, 0.0, 0.26700472831726074], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-0.15145523846149445, 25.0, 0.5526131987571716) (face-face (edge-edge) contact); contact 2 at (-0.17157518863677979, 25.0, 0.6252317428588867) (face-face (edge-edge) contact).

Exercise 2 (b) - Collision Details with e=0.0

Scenario 0: Default setup with no initial rotation.

For e=0, the assignment asks for first-contact properties only, so only collision 0 is reported.

Collision 0: t=2.504166666666686, collision point=[-0.004171311855316162, 25.0, 0.0], normal=[1.0, 0.0, -0.0], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-0.016671180725097656, 25.0, -0.5) (body 1 corner against body 0 face); contact 2 at (0.008328557014465332, 25.0, 0.5) (body 0 corner against body 1 face).

Scenario 1: Body 0 starts rotated by 30 degrees.

For e=0, the assignment asks for first-contact properties only, so only collision 0 is reported.

Collision 0: t=2.441666666666683, collision point=[0.23332858085632324, 25.0, -0.14322692155838013], normal=[1.0, 0.0, -0.0], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (0.23332858085632324, 25.0, -0.1617315709590912) (face-face (edge-edge) contact); contact 2 at (0.23332858085632324, 25.0, -0.12472227215766907) (face-face (edge-edge) contact).

Scenario 2: Body 0 starts at 60 degrees and body 1 starts at 30 degrees.

For e=0, the assignment asks for first-contact properties only, so only collision 0 is reported.

Collision 0: t=2.475000000000018, collision point=[-0.2613295912742615, 25.0, 0.1339745968580246], normal=[0.8660253882408142, 0.0, 0.5], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-0.25897935032844543, 25.0, 0.1299038976430893) (face-face (edge-edge) contact); contact 2 at (-0.2636798024177551, 25.0, 0.1380452960729599) (face-face (edge-edge) contact).

Scenario 3: Body 0 starts spinning at -5 rad/s, while body 1 is moved to x=-1.9, z=1.9 and starts from rest.

For e=0, the assignment asks for first-contact properties only, so only collision 0 is reported.

Collision 0: t=1.6999999999999953, collision point=[-2.403679370880127, 25.0, 0.8999999761581421], normal=[-0.0, 0.0, 1.0], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (-2.403160333633423, 25.0, 0.8999999761581421) (face-face (edge-edge) contact); contact 2 at (-2.404198408126831, 25.0, 0.8999999761581421) (face-face (edge-edge) contact).

Scenario 4: Body 0 starts spinning at +3 rad/s, and body 1 starts at -10 degrees.

For e=0, the assignment asks for first-contact properties only, so only collision 0 is reported.

Collision 0: t=2.5166666666666866, collision point=[0.12067129462957382, 25.0, -0.6252517104148865], normal=[0.9541522860527039, 0.0, 0.29932165145874023], contact mode=simultaneous-contacts-averaged, contacts before averaging=2.

Multiple contacts occurred at the same instant, so they were averaged into one collision point. Before averaging, the contact set was: contact 1 at (0.13062013685703278, 25.0, -0.6569657921791077) (face-face (edge-edge) contact); contact 2 at (0.11072244495153427, 25.0, -0.5935376286506653) (face-face (edge-edge) contact).

Exercise 3 - Momentum, Angular Momentum, and Energy

The table below lists values immediately before and immediately after the first collision in each scenario.

Restitution e=1.0

Scenario	Stage	P	L	K_trans	K_rot	K_total
0	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
0	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	2.859504595398903	7.140496293375321	10.000000888774224
1	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
1	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	2.6081303656101227	7.391870022851634	10.000000388461757
2	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
2	post	(-2.0, 0.0, 0.0)	(0.0, -2.999999523162842, 50.0)	7.4656572341918945	2.534342724375499	9.999999958567393
3	pre	(2.0, 0.0, 0.0)	(0.0, -4.3333330154418945, -50.0)	2.0	8.333333333333332	10.333333333333332
3	post	(2.0, 0.0, 0.0)	(0.0, -4.333333492279053, -50.0)	2.6483192145824432	7.685014132148712	10.333333346731155
4	pre	(-2.0, 0.0, 0.0)	(0.0, -1.0, 50.0)	10.0	3.0	13.0
4	post	(-2.0, 0.0, 0.0)	(0.0, -1.0, 50.0)	1.8578360080718994	11.142162735688643	12.999998743760543

Scenario 0: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is effectively conserved (up to numerical precision). Most of the lost translational energy is converted into rotation. (Delta_K_total=8.887742239949148e-07, Delta_K_trans=-7.140495404601097, Delta_K_rot=7.140496293375321).

Scenario 1: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is effectively conserved (up to numerical precision). Most of the lost translational energy is converted into rotation. (Delta_K_total=3.884617569838156e-07, Delta_K_trans=-7.391869634389877, Delta_K_rot=7.391870022851634).

Scenario 2: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 4.76837158203125e-07, 0.0). Total kinetic energy is effectively conserved (up to numerical precision). Most of the lost translational energy is converted into rotation. (Delta_K_total=-4.143260667888171e-08, Delta_K_trans=-2.5343427658081055, Delta_K_rot=2.534342724375499).

Scenario 3: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, -4.76837158203125e-07, 0.0). Total kinetic energy is effectively conserved (up to numerical precision). Part of the rotational energy is converted into translational motion. (Delta_K_total=1.3397823295235867e-08, Delta_K_trans=0.6483192145824432, Delta_K_rot=-0.6483192011846199).

Scenario 4: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is effectively conserved (up to numerical precision). Most of the lost translational energy is converted into rotation. (Delta_K_total=-1.2562394573478741e-06, Delta_K_trans=-8.1421639919281, Delta_K_rot=8.142162735688643).

Restitution e=0.0

Scenario	Stage	P	L	K_trans	K_rot	K_total
0	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
0	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	1.6694214083254337	1.7851240733438303	3.454545481669264
1	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
1	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	1.749851442873478	1.8479675057129086	3.5978189485863865
2	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0

Scenario	Stage	P	L	K_trans	K_rot	K_total
2	post	(-2.0, 0.0, 0.0)	(0.0, -2.999999761581421, 50.0)	3.324220657348633	0.6335856810938747	3.9578063384425075
3	pre	(2.0, 0.0, 0.0)	(0.0, -4.3333330154418945, -50.0)	2.0	8.333333333333332	10.333333333333332
3	post	(2.0, 0.0, 0.0)	(0.0, -4.333333492279053, -50.0)	2.162079803645611	7.973649159939757	10.135728963585368
4	pre	(-2.0, 0.0, 0.0)	(0.0, -1.0, 50.0)	10.0	3.0	13.0
4	post	(-1.9999998807907104, 0.0, 0.0)	(0.0, -1.0, 49.999996185302734)	4.192410558462143	5.915098889526714	10.107509447988857

Scenario 0: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 0.0, 0.0). Total kinetic energy drops, which is expected for an inelastic collision (e=0). Most of the lost translational energy is converted into rotation. (Delta_K_total=-6.545454518330736, Delta_K_trans=-8.330578591674566, Delta_K_rot=1.7851240733438303).

Scenario 1: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 0.0, 0.0). Total kinetic energy drops, which is expected for an inelastic collision (e=0). Most of the lost translational energy is converted into rotation. (Delta_K_total=-6.4021810514136135, Delta_K_trans=-8.250148557126522, Delta_K_rot=1.8479675057129086).

Scenario 2: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, 2.384185791015625e-07, 0.0). Total kinetic energy drops, which is expected for an inelastic collision (e=0). Most of the lost translational energy is converted into rotation. (Delta_K_total=-6.0421936615574925, Delta_K_trans=-6.675779342651367, Delta_K_rot=0.6335856810938747).

Scenario 3: momentum change Delta_P=(0.0, 0.0, 0.0), angular momentum change Delta_L=(0.0, -4.76837158203125e-07, 0.0). Total kinetic energy drops, which is expected for an inelastic collision (e=0). Part of the rotational energy is converted into translational motion. (Delta_K_total=-0.19760436974796391, Delta_K_trans=0.1620798036456108, Delta_K_rot=-0.3596841733935747).

Scenario 4: momentum change Delta_P=(1.1920928955078125e-07, 0.0, 0.0), angular momentum change Delta_L=(0.0, 0.0, -3.814697265625e-06). Total kinetic energy drops, which is expected for an inelastic collision (e=0). Most of the lost translational energy is converted into rotation. (Delta_K_total=-2.892490552011143, Delta_K_trans=-5.807589441537857, Delta_K_rot=2.915098889526714).